ENGINEERING DOCUMENTATION

System Release 2.0

Vidudhi Smart Residence & Hostel Living Platform

A complete, humanized guide explaining every module, underlying hardware and AI algorithms, and seamless backend cloud connections.

```
[Client SPA: React + Vite + Redux Toolkit]
    | (HTTPS / Bearer JWT Authorization Tokens / JSON Payload)
    ▼
[Backend API: Node.js + Express.js + Rate Limiters + Helmet Security]
    | (Mongoose ODM / TLS Encrypted Connection)
    ▼
[Database Cloud: MongoDB Atlas Cluster 'keerthana']
```

Target Platform

Web SPA (Desktop, Tablet, Mobile Fluid Responsive)

Database Target

MongoDB Atlas Cloud (Production Cluster: keerthana)

Core Security

JWT Authentication, Bcryptjs Salted Hashes, Brute-Force Rate Limiting

Hardware & Biometrics

GPS Geofencing, ResNet-34 Face AI, WebAuthn Fingerprint, HMAC SHA-256 QR

Chapter 1. Executive System Architecture

Vidudhi is an intelligent hostel and residential campus management platform. It replaces outdated paper logbooks and fragmented spreadsheets with a single, synchronized digital environment. Whether a student is checking in for attendance, requesting emergency medical aid, booking laundry slots, or obtaining gate clearance, the system coordinates every action across mobile and desktop devices.

1.1 The 3-Tier Distributed Architecture

The platform is architected into three distinct layers to ensure speed, security, and scalability:

LAYER	TECHNOLOGIES USED	ROLE & HUMAN EXPLANATION
Frontend Layer (Client SPA)	React 18, Vite 5, Redux Toolkit, React Router 6, Lucide Icons, Vanilla CSS Design Tokens.	The interactive portal loaded in the user's browser. It provides instantaneous page navigation without page reloads, role-specific navigation menus, fluid responsive layouts on mobile phones, and client-side biometric matching.
Backend API Layer (Server)	Node.js 18+, Express.js 4, Helmet Security, CORS, Express-Rate-Limit, Morgan Logger.	The central brain running in the cloud. It validates user requests, verifies digital signatures, calculates geofence distances, protects endpoints against password guessing (15 attempts/15 min limit), and serves REST API endpoints.
Database Layer (Persistence)	MongoDB Atlas Cloud (Cluster: `keerthana`), Mongoose 8 Object Data Modeling (ODM).	Enterprise cloud database cluster storing structured collections: Users, Rooms, Attendance Records, Complaints, Leave Passes, Visitors, Mess Menus, and System Audit Logs.

1.2 How the Frontend and Backend Talk to Each Other

Communication between the frontend and backend follows a strict, industry-standard REST API pattern:

1. **1. Secure Handshake (Login):** When a user logs in with their roll number or email and password, the frontend transmits an encrypted payload to `POST /api/auth/login`.
2. **2. Digital Passport (JWT Token):** The backend verifies the password using salted `bcryptjs` hashes. If valid, it returns a cryptographically signed **JSON Web Token (JWT)** containing the user's ID, role, and clearance level.
3. **3. Authenticated Requests:** For all subsequent requests, the frontend automatically attaches this token inside the HTTP header: `Authorization: Bearer <JWT_TOKEN>`.
4. **4. Gatekeeping Middleware:** Before executing any database read or write, backend middleware intercepts the request, verifies the signature, confirms user permissions (e.g. preventing a student from accessing admin controls), and only then executes the database query.

Chapter 2. Comprehensive Module Breakdown

Below is a humanized, detailed breakdown of all 20 modules that power the Vidudhi ecosystem.

1. Authentication & Role-Based Access Control

SECURITY & ACCESS

IN SIMPLE WORDS

This is the digital gatekeeper of the hostel. Instead of having confusing login screens with leaked test accounts, the login page is clean, elegant, and simple. It supports 5 distinct hostel roles: Student Resident, Chief Warden, System Admin, Mess Manager, and Campus Doctor.

FRONTEND TECH & UX

React form inputs with validation, Redux `authSlice` persisting session state, fluid 5-role toggle buttons, Google OAuth2 Identity Services SDK.

BACKEND API & DATABASE

`POST /api/auth/login`, `POST /api/auth/google`, `GET /api/auth/me`.
Backed by Mongoose `User` collection with `bcryptjs` salted passwords and brute-force rate limiter (15 requests/15 min).

2. Campus Overview & Command Center

ANALYTICS

IN SIMPLE WORDS

The first screen users see after logging in. For students, it highlights their room number, night roll-call status, outpass approvals, and dinner menu. For the Warden and Admin, it transforms into an executive telemetry cockpit tracking total residents, occupied beds, pending repairs, and active curfew clearances.

FRONTEND TECH & UX

Real-time stat widgets, SVG Donut occupancy graphs, quick navigation shortcuts, responsive CSS grid.

BACKEND API & DATABASE

`GET /api/admin/overview`, `GET /api/user/recently-accessed`.
Aggregates database totals across `Room`, `Complaint`, `Attendance`, and `Leave` collections.

3. Smart Attendance Desk (GPS Geofencing + Biometrics)

BIOMETRICS & AI

IN SIMPLE WORDS

Traditional night roll-call requires wardens to walk door-to-door with paper registers between 8:00 PM and 9:30 PM. Vidudhi automates this completely. Students verify attendance on their own phone using two tamper-proof checks: First, GPS confirms they are physically inside the hostel boundary (≤ 300 meters). Second, their camera verifies their face using an AI neural network or their phone's fingerprint sensor.

FRONTEND TECH & UX

HTML5 Geolocation API with Haversine distance formula; `@vladmandic/face-api` (128-dimensional ResNet-34 neural embedding inference with < 0.50 Euclidean distance); WebAuthn hardware biometric sensor API.

BACKEND API & DATABASE

`POST /api/attendance/check-in`, `GET /api/attendance/daily`, `PATCH /api/attendance/override/:id`.
Records GPS coordinates, face match distance, verification method, and timestamp in `Attendance` collection.

4. Room Allocation & Capacity Hub

FACILITIES

IN SIMPLE WORDS

Keeps track of every single room and bed across Block A and Block B. Shows exactly which beds are occupied, which are vacant, room maintenance status, and roommates. Wardens can reassign beds with a click, while students can see their room details and roommates.

FRONTEND TECH & UX

Visual room grid cards, bed allocation slot badges (Bed 1, 2, 3), occupancy filters (Fully Occupied, Partially Filled, Vacant).

BACKEND API & DATABASE

`GET /api/rooms`, `GET /api/rooms/:id`, `PATCH /api/rooms/allocate`.
Backed by `Room` schema with bed slots, resident roll number relations, and capacity counters.

5. Maintenance Grievance & Complaint Desk

SUPPORT

IN SIMPLE WORDS

When a fan breaks, a tap leaks, or the room Wi-Fi fails, students don't need to chase maintenance staff. They file a digital ticket with a priority tag (High, Medium, Low). Maintenance staff and wardens update the ticket as it moves from "Open" to "In Progress" to "Resolved". Students can only see their own room complaints to protect privacy.

FRONTEND TECH & UX

Ticket creation modal, category filter tabs (Electrical, Plumbing, Furniture, Cleanliness, Wi-Fi), high-contrast priority tags.

BACKEND API & DATABASE

`GET /api/complaints`, `POST /api/complaints`, `PATCH /api/complaints/:id/status`.
Backed by Mongoose `Complaint` model storing ticket IDs, resident IDs, room numbers, and status timestamps.

6. Leave Approvals & Cryptographic Turnstile QR Gatepasses

SECURITY & QR

IN SIMPLE WORDS

When a student needs to leave campus for the weekend or an emergency, they submit a digital leave application. The Chief Warden reviews and approves it. Once approved, the system generates a secure, tamper-proof QR gatepass on the student's phone. Security guards scan this QR code at the turnstile gate for instant clearance.

FRONTEND TECH & UX

Leave application form with date pickers; dynamic QR code rendering using `qrcode` and `jsqr`; turnstile gate scanner view.

BACKEND API & DATABASE

`GET /api/leave`, `POST /api/leave`, `PATCH /api/leave/:id/approve`, `POST /api/leave/verify/:id`.
Backed by `Leave` collection with cryptographic HMAC SHA-256 validation tokens and curfew timers.

7. Visitor Management & Security Desk

CAMPUS SECURITY

IN SIMPLE WORDS

Ensures safe campus entry for visiting parents, guardians, and official guests. Guards record the visitor's name, phone number, resident they are visiting, and purpose of visit. Generates a digital entry pass and logs exact entry and checkout timestamps.

FRONTEND TECH & UX

Searchable visitor table, quick check-in modal, visitor pass card preview, active guest badges.

BACKEND API & DATABASE

`GET /api/visitors`, `POST /api/visitors`, `PATCH /api/visitors/:id/checkout`.
Stored in `Visitor` collection with relation to visited student roll number and guard check-out timestamps.

8. Smart Laundry Hub & Garment Care

HOSTEL AMENITIES

IN SIMPLE WORDS

Prevents students from fighting over ground-floor washing machines or waiting in long queues. Students can check which machines (Machine 1, 2, 3) and steam iron stations are available, book a slot, and follow a 4-step progress bar (Requested → In Wash → Ironed → Ready for Pickup). All laundry is 100% covered under the annual hostel fee (¢0 charge).

FRONTEND TECH & UX

Machine availability cards, interactive booking modal, visual stepper timeline, $₹0$ fee notice badge.

BACKEND API & DATABASE

Synced through Redux store and user records, logging wash cycles, scheduled slots, and machine maintenance flags.

9. Mess Operations & Dining Feedback

DINING

IN SIMPLE WORDS

Displays the complete 7-day digital dining menu (Breakfast, Lunch, Evening Snacks, and Dinner). Students can rate each meal (1 to 5 stars) and submit feedback directly to the kitchen. A dedicated Mess Manager login allows kitchen staff to update tomorrow's menu and review student meal ratings without accessing student grades or personal data.

FRONTEND TECH & UX

Tabbed weekly schedule, interactive 5-star rating controls, special dish photo tags, add-on order toggles.

BACKEND API & DATABASE

`GET /api/mess/menu`, `POST /api/mess/feedback`, `PATCH /api/mess/menu/:day`.
Backed by `MessMenu` and `MessFeedback` collections storing daily meal options, ratings, and dietary comments.

10. Hostel Accounts & Official Fee Receipts

FINANCE

IN SIMPLE WORDS

A crystal-clear financial breakdown for students and parents. Shows hostel room rent, mess charges, and annual amenity fees. Students can review paid statements, check pending dues, and download official fee receipts with official transaction reference IDs.

FRONTEND TECH & UX

Ledger statements table, payment method selector, status pills (Paid, Overdue, Processing), printable receipt view.

BACKEND API & DATABASE

Integrated with resident account profile; calculates fee balances, receipt IDs, and transaction timestamps.

11. Campus Health & OPD Clinic

HEALTHCARE

IN SIMPLE WORDS

Hostel healthcare made private and accessible. Students can view campus doctor duty hours, check the live OPD token waiting queue, and book a consultation. Crucially, student medical privacy is strictly enforced: students only see their own prescriptions and current queue number, while doctors and wardens see clinical consultation records.

FRONTEND TECH & UX

Doctor schedule cards, real-time token queue number badge, private personal prescription ledger, symptom logger.

BACKEND API & DATABASE

Linked with user roll numbers and doctor staff profiles; logs token issuance, doctor remarks, and medicine clearances.

12. Emergency SOS Distress Monitor

LIFE SAFETY

IN SIMPLE WORDS

A life-safety emergency system. In any crisis (medical emergency, fire, physical distress), clicking the bright Tomato Red (#FF6347) SOS button instantly dispatches the resident's room number, GPS location, and distress message to the Chief Warden and campus ambulance. If the student's Wi-Fi drops, the SOS modal stores the distress trigger locally in offline queue and dispatches the moment connection returns.

FRONTEND TECH & UX

Emergency modal with high-contrast red styling; Web Audio API audible pulse chirp; offline-first localStorage queue listener; quick phone dialer (108 Ambulance, 101 Fire, 1091 Women Safety).

BACKEND API & DATABASE

Real-time distress telemetry dispatch; logs distress incident to Warden Command Center and database audit log.

13. AI Hostel Copilot

ARTIFICIAL INTELLIGENCE

IN SIMPLE WORDS

A 24/7 AI-powered residential assistant accessible via a dedicated page or floating chat widget. Students can ask questions in natural English like "What is the gate closing time on Saturday?", "What are the gym hours?", or "Can I borrow sports equipment?". The AI answers instantly based on the official college hostel rulebook, and even provides study focus tips and motivation.

FRONTEND TECH & UX

Puter.js AI SDK integration; floating chat drawer with minimize/maximize controls; instant keyword knowledge base fallback.

BACKEND API & DATABASE

`POST /api/chatbot/query`.
Hostel rulebook embeddings, conversational context memory, and safety filters.

14. Hostel Daily Forecaster (Simulation Engine)

ALGORITHMIC TELEMETRY

IN SIMPLE WORDS

A resource management tool for the Chief Warden. Rather than guessing how much electricity, water, or food the hostel will consume tomorrow, the warden adjusts sliders for occupancy and weather (e.g. 90% occupancy on a hot summer day). The engine calculates expected electricity (kWh), water usage (Liters), mess meal portions (plates), and generator fuel needs, saving thousands of rupees in wasted resources.

FRONTEND TECH & UX

Interactive parameter sliders (occupancy, temperature, study hours); visual projected telemetry cards; resource breakdown bars.

BACKEND API & DATABASE

`POST /api/simulation/forecast`, `GET /api/simulation/scenarios`.
Mathematical demand modeling algorithms calculating utility consumption matrices based on historical occupancy.

15. Digital Campus Map & Navigation

NAVIGATION

IN SIMPLE WORDS

Helps junior students and visitors orient themselves within the campus. An interactive visual map displays Block A, Block B, the central dining mess, sports ground, health clinic, laundry room, library, and the security main gate with live opening hours and status tags.

FRONTEND TECH & UX

Vector SVG campus layout with interactive facility nodes, hover tooltips, status pills (Open, Closed, Restricted).

BACKEND API & DATABASE

Integrated with facility operating hours and campus coordinates config (`HOSTEL_LAT`, `HOSTEL_LNG`).

16. Facilities & Sports Hub

RECREATION

IN SIMPLE WORDS

Manages sports gear and fitness gym access. Students can borrow badminton sets, cricket kits, basketballs, or table tennis gear with automatic checkout countdown timers (2-4 hours). They can also reserve 45-minute campus gym workout slots to prevent overcrowding in the weight room.

FRONTEND TECH & UX

Gear catalog cards, live borrow timer counters, gym workout slot reservation toggles, equipment condition badges.

BACKEND API & DATABASE

`GET /api/equipment`, `POST /api/equipment/borrow`. Backed by `Equipment` collection tracking available inventory count, borrowed user roll numbers, and return dues.

17. Lost & Found Campus Registry

COMMUNITY

IN SIMPLE WORDS

A digital notice board for lost and found campus belongings. If a student finds keys, an ID card, a watch, or a charger in the study hall, they can post a photo and location. The rightful owner can claim it by verifying identifying details with the finder or warden.

FRONTEND TECH & UX

Card feed with photo previews, item category tags, report modal with image attachment, status filters (Reported, Claimed).

BACKEND API & DATABASE

Stored in database collection linking reporting student, item description, location found, and resolved timestamp.

18. Campus Notice Board & Circulars

ANNOUNCEMENTS

IN SIMPLE WORDS

Official broadcast channel for Chief Warden announcements, holiday hostel closures, maintenance schedules, and curfew alerts. Notices appear instantly with high-contrast priority tags. An unread badge counter in the top bar ensures students never miss a critical notice.

FRONTEND TECH & UX

Chronological circular cards, priority badges (Urgent, Notice, General), notification count badge on bell icon.

BACKEND API & DATABASE

`GET /api/notifications`, `POST /api/notifications`. Mongoose `Notification` schema storing publisher ID, broadcast scope, title, message body, and publish date.

19. User Directory & Administrative Controls

ADMINISTRATION

IN SIMPLE WORDS

The administrative backbone of the hostel. Allows administrators to enroll new students, register staff, update room assignments, and deactivate graduated residents. Strict role separation ensures that the Warden has read-only lookup access, while full editing and deleting powers are restricted to System Administrators.

FRONTEND TECH & UX

Searchable user data table, role filter dropdown, add user modal with input validation, lock indicators for non-admins.

BACKEND API & DATABASE

`GET /api/admin/users`, `POST /api/admin/users`, `DELETE /api/admin/users/:id`.
Protected by admin authorization middleware; directly performs CRUD operations on the Mongoose `User` collection.

20. System Personalization, Theming & Multi-Language

PERSONALIZATION

IN SIMPLE WORDS

Provides an inclusive, accessible user experience tailored to every resident. Users can switch between dark and light modes, choose modern leafy-green accent palettes, pick button styles (Modern Solid or Clean Outline), customize typography fonts, and switch languages between English, Tamil, and Hindi.

FRONTEND TECH & UX

CSS Custom Properties (tokens), HTML `data-font` and `dir` attributes, Redux `uiSlice`, local storage persistence.

BACKEND API & DATABASE

`PATCH /api/user/language`.
Saves the user's preferred interface language and display settings in their database profile.

Chapter 3. Complete Backend API Route Map

Below is the complete catalogue of REST API endpoints exposed by the Node.js/Express server and consumed by the React frontend:

METHOD	ENDPOINT URL	AUTHORIZED ROLE	DATABASE OPERATION / ACTION
POST	/api/auth/login	Public (Rate limited 15/15m)	Validates credentials, returns signed JWT
POST	/api/auth/google	Public	Verifies Google OAuth2 ID token
GET	/api/auth/me	All Logged-in Roles	Fetches current user profile from JWT session
GET	/api/admin/overview	Warden, Admin	Aggregates room capacity, leaves, complaints
GET	/api/admin/users	Warden, Admin	Retrieves directory of students and staff
POST	/api/admin/users	Admin Only	Enrolls new resident with encrypted password
DELETE	/api/admin/users/:id	Admin Only	Removes resident record and frees allocated bed
POST	/api/attendance/check-in	Student	Validates GPS coordinates & FaceID vector
GET	/api/attendance/daily	Warden, Admin	Returns roll-call roster for current date
PATCH	/api/attendance/override/:id	Warden Only	Manual override for medical or duty absence
GET	/api/rooms	Student, Warden, Admin	Returns room list, occupancy, and bed slots
PATCH	/api/rooms/allocate	Warden, Admin	Assigns resident roll number to specific bed
GET	/api/complaints	Student (own), Warden (all)	Fetches maintenance tickets with status tags
POST	/api/complaints	Student	Submits a new maintenance grievance
PATCH	/api/complaints/:id/status	Warden, Admin	Updates ticket to In Progress or Resolved
GET	/api/leave	Student (own), Warden (all)	Lists outpass and overnight leave applications
POST	/api/leave	Student	Applies for digital leave with parent intimation
PATCH	/api/leave/:id/approve	Warden Only	Approves leave and issues HMAC QR gatepass
POST	/api/leave/verify/:id	Warden, Security Guard	Scans and clears QR token at gate turnstile

METHOD	ENDPOINT URL	AUTHORIZED ROLE	DATABASE OPERATION / ACTION
GET	`/api/visitors`	Warden, Admin	Fetches active campus guest log
POST	`/api/visitors`	Warden, Security Guard	Registers visiting parents or guests
GET	`/api/mess/menu`	All Roles	Returns 7-day breakfast, lunch, dinner schedule
POST	`/api/mess/feedback`	Student	Logs 5-star meal ratings and kitchen comments
POST	`/api/simulation/forecast`	Warden, Admin	Calculates electricity, water, meal predictions
POST	`/api/chatbot/query`	Student	NLP search over official hostel rulebook
GET	`/api/notifications`	All Roles	Fetches official circulars and curfew updates
POST	`/api/notifications`	Warden, Admin	Broadcasts new circular to all residents